

Sensing

Learning Objectives

- Use Sensing blocks to detect interactions
- Detect mouse position, key presses, and sprite collisions
- Create interactive projects using sensing

Engage (5-7 minutes)

Think of a scarecrow in a field that reacts when birds come near. Sensing blocks let your sprites detect what is happening around them—whether something is touching them, the mouse is moving, or a key is pressed. Today we give our sprites senses like real-world objects.

Explore (10-12 minutes)

Try “touching mouse-pointer?” to make a sprite follow your cursor. Use “distance to” to measure how far sprites are apart. Press keys and use “key pressed?” to trigger actions. Ask questions with “ask and wait” to get user input and store answers in a variable.

Explain (10-12 minutes)

Sensing blocks let sprites detect their environment. You can check if sprites touch each other, sense mouse and key input, get user responses with “ask,” and measure distances and time. These blocks make your projects interactive by responding to what players do.

Elaborate (8-10 minutes)

Create a game where a cat sprite catches mice—use “touching” to detect when they meet. Build a quiz where the sprite asks questions and checks answers. Make a drawing tool where the pen follows the mouse using sensing. Like a village well that tells you how much water is there, sensing tells your sprites about their world.

Evaluate (5-8 minutes)

Exit ticket: Name three things Sensing blocks can detect. How would you make a sprite change color when the mouse touches it? What does the “ask” block do and where is the answer stored?